J R SARVES SRIRANGAN

AI Engineer / Full Stack Developer

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PROFESSIONAL SUMMARY

Computer Science student specializing in Artificial Intelligence, Machine Learning, Data Science, and Full Stack Development, with hands-on experience developing AI applications, healthcare software, computer vision systems, and data-driven solutions. Skilled in Python, JavaScript, React.js, Node.js, Express.js, SQL, Machine Learning, Computer Vision, and REST API development, with experience delivering production-ready web applications and database-driven systems. Also freelances as a developer, building websites, software applications, and Flutter-based mobile applications. Actively strengthening core problem-solving skills through consistent practice of Data Structures & Algorithms on LeetCode.

TECHNICAL SKILLS

Languages: Python, Java, C, C++

Web Development: HTML, CSS, JavaScript, React.js, Node.js, Express.js, REST APIs, JSON, CRUD Operations, Authentication, Responsive Web Design, Flutter

Backend & Databases: SQL, MySQL, PostgreSQL, MongoDB, Database Design, Database Administration, API Integration, Software Architecture

AI & Data Analytics: Machine Learning, Deep Learning (CNNs, RNNs), Computer Vision (OpenCV), Scikit-learn, TensorFlow, PyTorch, Generative AI, LLM Integration, Prompt Engineering, Pandas, NumPy, Advanced Excel, Power BI, Tableau, Matplotlib, Seaborn, Data Cleaning, EDA, Statistical Analysis, A/B Testing, Data Storytelling

Blockchain: MetaMask, Web3, ERC-8004, x402

Tools & Cloud: Git, Git Workflow, GitHub, Linux, VS Code, Jupyter Notebook, Android Studio, Streamlit, Vercel

PROFESSIONAL EXPERIENCE

Full Stack Developer & Database Administrator — Annavi Trust **Jun 2026 – Aug 2026**

Freelance · Chennai, India

- Developed a clinic management platform comprising 3+ integrated modules and 20+ database tables, used to digitize patient records, inventory management, and billing workflows and replace manual record-keeping for day-to-day operations.
- Built patient registration, visit history, prescription tracking, and searchable EMR workflows, along with inventory management for stock procurement, movement, and expiry tracking.
- Developed billing workflows integrating consultation charges and pharmacy sales with automatic inventory updates.
- Designed and normalized a relational MySQL database consisting of multiple interconnected tables with foreign key constraints, indexing, and optimized SQL queries for efficient data retrieval.
- Developed RESTful APIs supporting CRUD operations for patients, inventory, billing records, and administrative workflows using Node.js and Express.js, integrated with a responsive React.js frontend.
- Deployed the production application and website while maintaining application availability and database administration.

Data Scientist — Atsuya Technologies **Jun 2026 (1 Month)**

Internship · Chennai, Tamil Nadu, India · On-site

- Worked on predictive maintenance, fault diagnostics, and performance analytics for 1000+ HVAC systems using Machine Learning and Data Science, developing data-driven methods to identify refrigerant faults, short cycling, icing, and other operational anomalies.
- Applied OpenCV, YOLO, and digital image processing techniques to an industrial oil seed inspection project, building a custom filtering pipeline to improve defect detection in low-texture, poorly-imaged regions.
- Debugged and resolved a production race condition in the AONE-Eye occupancy detection task scheduler, improving pipeline startup reliability.

EDUCATION

B.Tech — Computer Science and Engineering Expected: 2028
*SRM Institute of Science and Technology • CGPA: **9.74***

Foundation Degree — Data Science and Programming Dec 2025
Indian Institute of Technology, Madras (IITM) • Chennai, India

TECHNICAL PROJECTS

Medical Report Analysis System (Hackathon Project)

React.js, Node.js, LLM, JWT Authentication, MetaMask, ERC-8004, x402
GitHub | Live Demo | Video Demo

- Built a full-stack AI application supporting PDF and image medical report analysis using Groq LLM with OCR-based document parsing and conversational querying.
- Developed an AI-powered conversational assistant that answers personalized questions based on uploaded medical reports using contextual memory and prompt engineering.
- Engineered a pipeline to read medical reports from PDF and image inputs, applying OCR and vision-based workflows to extract and structure report data.
- Implemented JWT-based authentication, MetaMask login, ERC-8004 AI agent identity, and x402 blockchain payment workflow via secure REST APIs for decentralized AI service access.
- Deployed the application on Vercel with CI/CD using GitHub, pairing a responsive React UI with a Node.js backend to deliver an end-to-end AI-powered healthcare experience.

AI Car Insurance Claim Analyzer

Python, Streamlit, OpenCV, Scikit-learn, Random Forest, Computer Vision
GitHub | Live Demo

- Built a machine learning-based insurance claim prediction system using Random Forest models for claim approval classification and claim amount estimation.
- Developed an OpenCV image processing pipeline for vehicle damage detection using contour analysis, edge detection, thresholding, and severity classification.
- Developed and deployed an interactive Streamlit application supporting real-time image upload, vehicle damage assessment, and ML-powered insurance claim prediction.
- Implemented business logic for repair cost estimation, depreciation, insurance coverage, deductibles, and customer liability calculations.
- Generated and utilized synthetic insurance datasets with Pandas and NumPy for model training, feature engineering, and prediction.

CERTIFICATIONS

- NPTEL — Data Structures and Algorithms
- SIDH, Reliance Foundation Skilling Academy — Artificial Intelligence
- SIDH, Reliance Foundation Skilling Academy — IoT Network Specialist
- Deloitte Australia — Data Analytics Job Simulation (Forage)
- Quantium — Data Analytics Job Simulation (Forage)

ACHIEVEMENTS

- Solved 120+ LeetCode problems across Data Structures and Algorithms using Python and Java View Profile
- Competitive Programming Lead in Logic Play SRM Institute technical club
- Secured 3rd Place in hackathon hosted by SSN College of Engineering, collaborating with a team to develop an innovative solution through technical problem-solving and teamwork.
- Selected for the live evaluation round at Amypo Hackathon, placing in the top 200 of 1000+ registered teams
- Earned 5 MongoDB Skill Badges from MongoDB University, including RAG, AI agents, AI data strategy, and vector search